

Software Requirements Specification

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****Software Requirements Specification (SRS)****

1. Introduction

This Software Requirements Specification (SRS) document describes the requirements for the B09 application, which is a web and mobile-based platform designed to promote eco-friendly activities and provide a leaderboard for users. The application is built using a combination of front-end and back-end technologies, including Vercel, Vite, React, and Express.js, with a MongoDB database.

2. Purpose

The purpose of this SRS is to outline the functional and non-functional requirements of the B09 application, including the system architecture, risks, constraints, and future scope. This document will serve as a guide for the development team, stakeholders, and users to ensure that the application meets the required standards and functionality.

3. Scope

The scope of this SRS includes the following components:

- * Front-end: React-based web application and mobile application (using React Native)
- * Back-end: Express.js API with MongoDB database
- * API Calls: Integration with various routes for user registration, activity completion, and leaderboard retrieval
- * Database Models: User, Activity, Badge, Image, and Leaderboard models

4. System Overview

The B09 application is a web and mobile-based platform that allows users to register, complete eco-friendly activities, and view their progress on a leaderboard. The application consists of the following modules:

- * User Registration: Allows users to register and create an account
- * Activity Completion: Allows users to complete eco-friendly activities and earn badges
- * Leaderboard: Displays the top users based on their activity completion and badge earnings
- * Activities Screen: Displays a list of available activities for users to complete

5. Functional Requirements

5.1 User Registration

- * The system shall allow users to register and create an account
- * The system shall validate user input data (e.g., email, password)
- * The system shall store user data in the MongoDB database

5.2 Activity Completion

- * The system shall allow users to complete eco-friendly activities
- * The system shall update user activity completion status in the database
- * The system shall award badges to users based on their activity completion

5.3 Leaderboard

- * The system shall display the top users based on their activity completion and badge earnings
- * The system shall retrieve leaderboard data from the database
- * The system shall update the leaderboard in real-time

5.4 Activities Screen

- * The system shall display a list of available activities for users to complete
- * The system shall retrieve activity data from the database
- * The system shall allow users to filter activities by category

6. Non-Functional Requirements

6.1 Performance

- * The system shall respond to user input within 2 seconds
- * The system shall handle a minimum of 100 concurrent users

6.2 Security

- * The system shall use HTTPS protocol for secure data transmission

- * The system shall validate user input data to prevent SQL injection attacks

6.3 Usability

- * The system shall provide an intuitive and user-friendly interface
- * The system shall provide clear instructions for user registration and activity completion

7. Architecture

The B09 application will be built using a microservices architecture, with the following components:

- * Front-end: React-based web application and mobile application (using React Native)
- * Back-end: Express.js API with MongoDB database
- * API Gateway: Vercel API Gateway for routing and security

8. Risks

The following risks have been identified:

- * Data security risks due to SQL injection attacks or data breaches
- * Performance issues due to high traffic or inadequate infrastructure
- * Integration risks due to incompatible API versions or data formats

9. Constraints

The following constraints have been identified:

- * Limited budget for infrastructure and development
- * Tight deadline for project completion
- * Limited resources for testing and quality assurance

10. Future Scope

The following features are planned for future development:

- * Integration with social media platforms for sharing activity completion and badge earnings

- * Development of a gamification module to encourage user engagement
- * Expansion of the activity database to include more eco-friendly activities and categories

Recent development updates have included the following:

- * Updation of vercel.json to improve routing and security
- * Addition of vercel.json to enable Vercel API Gateway
- * Merge of branch 'main' of <https://github.com/HACKWAVE2025/B09> to integrate latest changes
- * Fixing of Vercel routing for Single-Page Application (SPA)
- * Vite changes to improve front-end performance
- * DailyQuest automatic task completion
- * Index redirects to improve user experience
- * Backend deploy to production environment
- * Activities Drawer EcoCrush Task Done in Mobile to improve user engagement

The repository context includes the following files and folders:

- * backend/models/Activity.js: MongoDB model for Activity
- * backend/server.js: API call for activities
- * backend/routes/userRoutes.js: Express route for user registration and activity completion
- * backend/models/Badge.js: MongoDB model for Badge
- * backend/models/User.js: MongoDB model for User
- * mobile/app/(tabs)/Activities.tsx: React component for Activities Screen
- * backend/models/Image.js: MongoDB model for Image
- * frontend/src/pages/ActivitiesPage.jsx: API call for activities
- * backend/routes/leaderboardRoutes.js: Express route for leaderboard retrieval

Note: This SRS document is subject to change and updates as the project progresses.